Section: **3.2 – MECHANICALLY FASTENED JOINTS**

Chapter: **3216**

Title: **FASTENER LOADS IN LONGITUDINAL JOINTS OF MULTIPLE PLATES**

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1 INTRODUCTION

This chapter provides the methodology to be considered to calculate the loads transferred by the fasteners in a longitudinal joint of several plates.

1.1 METHOD-SUMMARY

The joint will be calculated considering only unidimensional aspects. This means that secondary bending effects as well as out of plane loads and deformations are not going to be considered.

The following hypotheses and considerations are going to be made:

- The external loads can be axial or shear loads (in plane) applied at the ends of the plates. Of course, they must be in equilibrium.

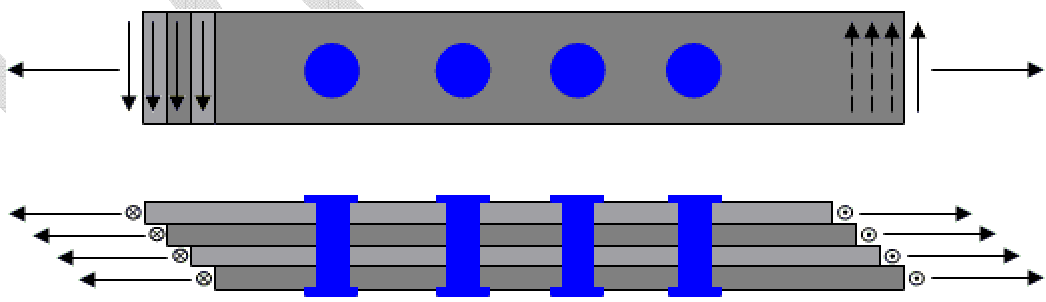

Figure 1: External loads

- Fasteners are installed without clearance and they will work in pure shear, transferring axial and shear load between the plates they are joining.

- Nonlinear effects, like material plasticity, will not be taken into account.

The fastener loads are calculated by using a displacement compatibility method.

After that, the load state of each plate along its longitudinal coordinate can be calculated considering only the external loads as well as the fasteners loads applied on it.

And finally, reserve factors for fasteners and plates can be calculated accounting for the load state and the allowable values of the corresponding materials.

1.2 SCOPE AND LIMITATIONS

The objective is to calculate the load transferred by each fastener and the load state along the longitudinal coordinate of each plate of the joint.

Reserve factors for fasteners and plates can be calculated accounting for the methodology described in other chapters of this Stress Methods Manual.

As already pointed previously:

- Only quasi-unidimensional configurations are covered.
- Nonlinear effects, like material plasticity, will not be taken into account.
- External loads can be only in-plane loads (axial or shear) applied at the ends of the elements.

2 METHOD DESCRIPTION

A general configuration of a multi-element joint can include “Ne” elements (plates) joined by “Nr” single fasteners.

2.1 GENERAL METHOD

<u>Kind of loading:</u>

The following equations and calculation process are going to be described only for axial loads combined with and increment of temperature.

For shear loads, the process is totally analogue as for axial loads. Then both cases can be solved separately.

After that, both solutions in terms of fastener loads and load state of the elements can be superimposed because of the linear nature of the analysis.

<u>Stiffness of fasteners:</u>

The stiffness of a fastener joining two adjacent elements can be calculated by means of the Huth’s formula, provided in chapter 3210.

<u>Group of fasteners:</u>

Sometimes, several fasteners can be in the same section or very near each other. In such a case a group of “n” fasteners can be associated to a single fastener with the following characteristics:

- Stiffness: The total amount of the stiffness of each fastener:

$$Total\ Stiffness = \sum_{i=1}^{n} Stiffness(i)$$

- Location: Calculated by ponderation of coordinates:

$$X = \frac{1}{Total\ Stiffness} \cdot \sum_{i=1}^{n} x(i) \cdot Stiffness(i)$$

<u>Stiffness of a part of an element between two consecutive fasteners:</u>

Since the part of an element between two consecutive fasteners (or groups of fasteners) can be divided in several segments and each one can vary in dimensions and properties, the flexibility of this part have to be calculated by integrating the flexibility along the axial coordinate:

$$Flexibility = \int_{x(i)}^{x(i+1)} \frac{dx}{E \cdot w \cdot t}$$

Note: For shear loads the expression will be the same, but using the shear modulus "*G*" instead of the elastic modulus "*E*".

Finally, the global stiffness of the part segment will be the inverse of the global flexibility:

$$Stiffness = \frac{1}{Flexibility}$$

<u>Forces and displacements in an element between two consecutive fasteners:</u>

The following figure shows the nomenclature for the <u>element "k"</u>, <u>segment "i"</u>, between fasteners "i-1" and "i":

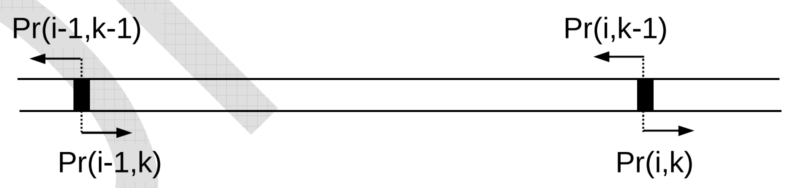

Figure 2: Segment of an element between two consecutive fasteners

Pr(i,k) is the force of fastener "i" acting on element "k" due to the load transference between elements "k" and "k+1".

The internal force of the of the element "k", segment "i" can be calculated by applying the equilibrium of forces for the element, from the left end up to this segment:

$$F(i,k) = P_{right}(k) + \sum_{j=i}^{Nr} [Pr(j,k) - Pr(j,k-1)]$$

Equation 1: Internal force in an element

Where $P_{rigth}(k)$ Axial load applied at the right end of the element "k"

> Note: For shear loads the expression will be the same, but using *Qright* (shear load applied at the right end of the element "k") instead of *Pright*.

The relative displacement between points "i-1" and "i" of the element "k" is given be the following expression:

$$\delta(i,k) - \delta(i-1,k) = \frac{F(i,k)}{K(i,k)} + \Delta T \cdot L(i) \cdot \alpha(i,k)$$

Equation 2: Relative displacement between points

Where
- $\delta(i,k)$ Displacement of the point at fastener "i" of element "k".
- $F(i,k)$ Internal force of the element "k" between fasteners "i-1" and "i".
- $K(i,k)$ Stiffness of the element "k", segment "i" (between fasteners "i-1" and "i").
- ΔT Increment of temperature over the assembly temperature.
- $L(i)$ Length of segment "i". $L(i) = X(i) – X(i-1)$
- $\alpha(i,k)$ Thermal expansion coefficient of the element "k", segment "i".

> Note: The increment of temperature will be considered inside of axial load case. For shear load case is $\Delta T = 0$.

Displacements at fastener locations:

The relative displacement between two points at two consecutive elements "k" and "k+1" joined by the fastener "i" is:

$$\delta(i,k+1) - \delta(i,k) = \frac{Pr(i,k)}{S(i,k)}$$

Equation 3: Relative displacement between elements at a fastener location

Where
- $Pr(i,k)$ Load transferred from element "k" to element "k+1 by fastener "i".
- $S(i,k)$ Stiffness of the fastener "i" between elements "k" and "k+1".

> Note: This expression is also valid for the shear load case.

Equilibrium of loads in an element:

The equilibrium of loads in an element "k" is formulated by means of the following equation:

$$P_{left}(k) - P_{right}(k) = \sum_{j=1}^{Nr} [Pr(j,k) - Pr(j,k-1)]$$

Equation 4: Loads equilibrium in an element

> Note: For shear loads the expression will be the same, but using *Qleft* and *Qright* instead of *Pleft* and *Pright*.

Calculation of fastener loads:

The total number of fastener loads is: Nr · (Ne-1). They can be calculated from all the above expressions.

For this purpose, it is necessary to remove de displacements “δ” from all the equations. Applying Equation 2: for elements “k+1” and “k”, between fasteners “i-1” and “i”, we have:

$$\delta(i,k)-\delta(i-1,k)=\frac{F(i,k)}{K(i,k)}+\Delta T\cdot L(i)\cdot\alpha(i,k)$$

$$\delta(i,k+1)-\delta(i-1,k+1)=\frac{F(i,k+1)}{K(i,k+1)}+\Delta T\cdot L(i)\cdot\alpha(i,k+1)$$

Subtracting both expressions and substituting the differences of displacements of both elements at each fastener point by Equation 3:

$$-\frac{Pr(i,k)}{S(i,k)}+\frac{Pr(i-1,k)}{S(i-1,k)}-\frac{F(i,k)}{K(i,k)}+\frac{F(i,k+1)}{K(i,k+1)}=\Delta T\cdot L(i)\cdot[\alpha(i,k)-\alpha(i,k+1)]$$

Finally, substituting the internal forces in elements according to Equation 1:

$$-\frac{Pr(i,k)}{S(i,k)}+\frac{Pr(i-1,k)}{S(i-1,k)}-$$

$$\frac{1}{K(i,k)}\cdot\sum_{j=i}^{Nr}[Pr(j,k)-Pr(j,k-1)]+\frac{1}{K(i,k+1)}\cdot\sum_{j=i}^{Nr}[Pr(j,k+1)-Pr(j,k)]=$$

$$-\frac{P_{right}(k+1)}{K(i,k+1)}+\frac{P_{right}(k)}{K(i,k)}+\Delta T\cdot L(i)\cdot[\alpha(i,k)-\alpha(i,k+1)]$$

Equation 5: Displacement compatibility

The above expression represents a set of (Nr – 1) · (Ne – 1) equations, where:

- i = 2, 3, …, Nr
- k = 1, 2, …, Ne – 1

With equations Equation 4: and equations Equation 5:, we have a system of Nr · (Ne - 1) equations, linearly independent, to determine the Nr fastener loads between the Ne elements.

2.2 Case of non-continuous panels

A plate can be split in two or more subpanels. All of them can be considered as the same element, but with null thickness in the discontinuity, as shown in next example:

Figure 3: Example of non-continuous panels

In the above figure, the thickness between points 5 and 6, and between points 9 and 10, is null. This means that the segments between consecutive fasteners including one of the above discontinuities have Stiffness = 0.

This case can be solved by applying the general method except when the displacement compatibility equation (Equation 5:) cannot be applied because K(i,k) = 0 or K(I,k+1) = 0.

In those cases where the general equation cannot be applied, it has to be substituted by another one, depending on the case:

<u>Case K(i,k) > 0 and K(i,k+1) > 0:</u>

The displacement compatibility equation (Equation 5:) is valid:

$$-\frac{Pr(i,k)}{S(i,k)}+\frac{Pr(i-1,k)}{S(i-1,k)}-$$

$$\frac{1}{K(i,k)}\cdot\sum_{j=i}^{Nr}[Pr(j,k)-Pr(j,k-1)]+\frac{1}{K(i,k+1)}\cdot\sum_{j=i}^{Nr}[Pr(j,k+1)-Pr(j,k)]=$$

$$-\frac{P_{right}(k+1)}{K(i,k+1)}+\frac{P_{right}(k)}{K(i,k)}+\Delta T\cdot L(i)\cdot[\alpha(i,k)-\alpha(i,k+1)]$$

<u>Case K(i,k) > 0 and K(i,k+1) = 0:</u>

The two terms containing K(i,k1) must be removed from the compatibility equation because the sum of both is zero (due to loads equilibrium).

$$-\frac{Pr(i,k)}{S(i,k)}+\frac{Pr(i-1,k)}{S(i-1,k)}-\frac{1}{K(i,k)}\cdot\sum_{j=i}^{Nr}[Pr(j,k)-Pr(j,k-1)]=$$

$$\frac{P_{right}(k)}{K(i,k)}+\Delta T\cdot L(i)\cdot[\alpha(i,k)-\alpha(i,k+1)]$$

<u>Case K(i,k) = 0 and K(i,k+1) = 0:</u>

In this case the compatibility equation has to be substituted by the corresponding to load equilibrium between the left end of the element and the discontinuity, as follows:

$$P_{left}(k)=\sum_{j=1}^{i-1}[Pr(j,k)-Pr(j,k-1)]$$

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
X	mm	X coordinate of geometric points
W	mm	Element Width at X coordinates
t	mm	Element Thickness at X coordinates
E	MPa	Material Elastic modulus
G	MPa	Material Shear modulus
$\mu 1$		Material Poisson's ratio in axial direction
$\mu 2$		Material Poisson's ratio in transversal direction
α	$^{o}C^{-1}$	Thermal expansion coefficient
Fsu	MPa	Ultimate shear stress
Ftu	MPa	Ultimate tensile stress
Fbru	MPa	Ultimate bearing stress (only metallic)
A	$\mu\varepsilon$	Allowable by-pass strain around a fastened attachment
B	MPa	Allowable bearing stress for the tension-through-the-hole failure
C	MPa	Allowable bearing stress for the bearing failure
Pleft	N	Axial load applied at the left end of the elements
Pright	N	Axial load applied at the right end of the elements
Qleft	N	Shear load applied at the left end of the elements
Qright	N	Shear load applied at the right end of the elements
ΔT	^{o}C	Increment of temperature

REFERENCES

N/A

SOFTWARE

Program	***Version***	***Description***

Table 1: Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.1 12/06/2020	Darío Fernández	First issue	All pages

Table 2: Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEASP-TL3)	Gustavo Baños DD/MM/AAAA	Florian Glock DD/MM/AAAA

Table 3: Approval signatures table for last revision of this chapter